

Executive Summary

Instagram Data Analysis for Engagement Optimization

Objective

The objective of this project is to analyze Instagram engagement data to identify optimal posting times, effective hashtag usage, and content strategies that maximize audience interaction and support follower growth.

Methodology

Instagram datasets containing posts, likes, follows, and hashtags were analyzed using Python. Data preprocessing included merging multiple datasets, extracting posting hours, and calculating engagement metrics based on likes and interactions.

Exploratory Data Analysis was performed to evaluate posting schedules, hashtag popularity, engagement distribution, and content performance.

Key Findings

- Certain posting hours consistently generate higher average engagement than others.
- A small group of hashtags contributes significantly to post visibility.
- Engagement varies across different posting times and content categories.
- High-performing posts share common characteristics such as optimal timing and strategic hashtag usage.

Business Insights

An optimized posting schedule combined with consistent branding and targeted hashtags can significantly improve audience reach and engagement. Data-driven content planning enables better allocation of marketing resources and supports long-term follower growth.

Recommendation

Develop a structured weekly content calendar that schedules posts during peak engagement hours while balancing educational, promotional, and interactive content. Regular performance monitoring should be used to refine the strategy over time.

Instagram Growth Strategy for Alfido Tech

Recommended Weekly Posting Plan

Day	Content
Monday	Educational AI or technology tips
Tuesday	Product feature demonstration
Wednesday	Customer success story or testimonial
Thursday	Industry trends and insights
Friday	Behind-the-scenes or team culture
Saturday	Interactive poll, quiz, or Q&A
Sunday	Weekly recap or motivational post

Five Engagement Strategies

1. Publish content during peak engagement hours identified through analysis.
2. Use a consistent set of high-performing hashtags relevant to technology and AI.

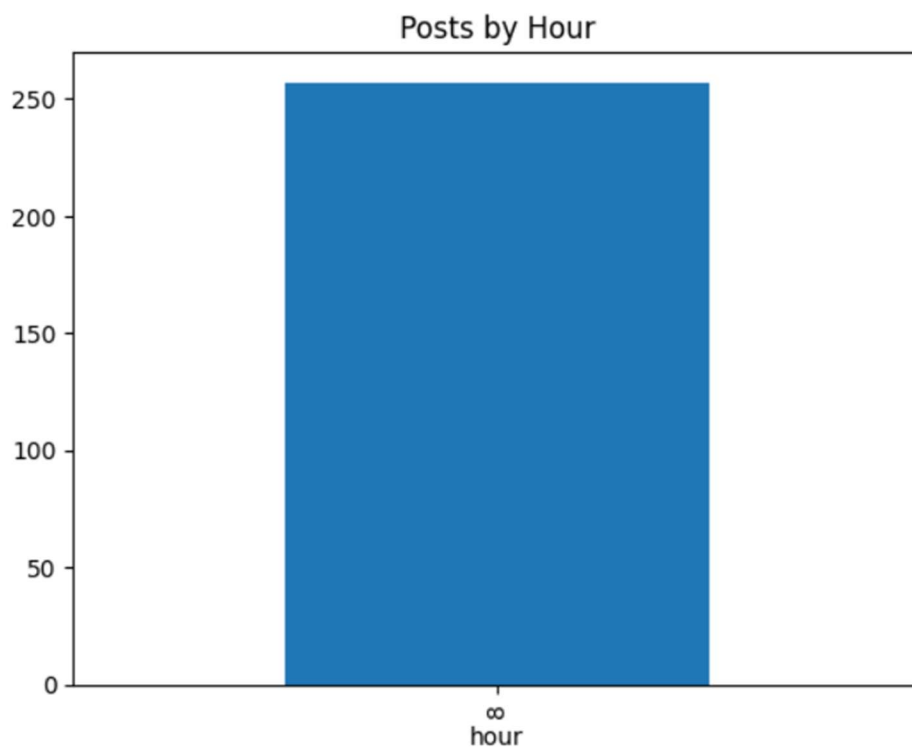
- 3. Increase audience interaction through polls, questions, and carousel posts.**
- 4. Mix educational and promotional content to provide value while building brand awareness.**
- 5. Monitor engagement metrics weekly and adjust posting schedules based on performance trends.**

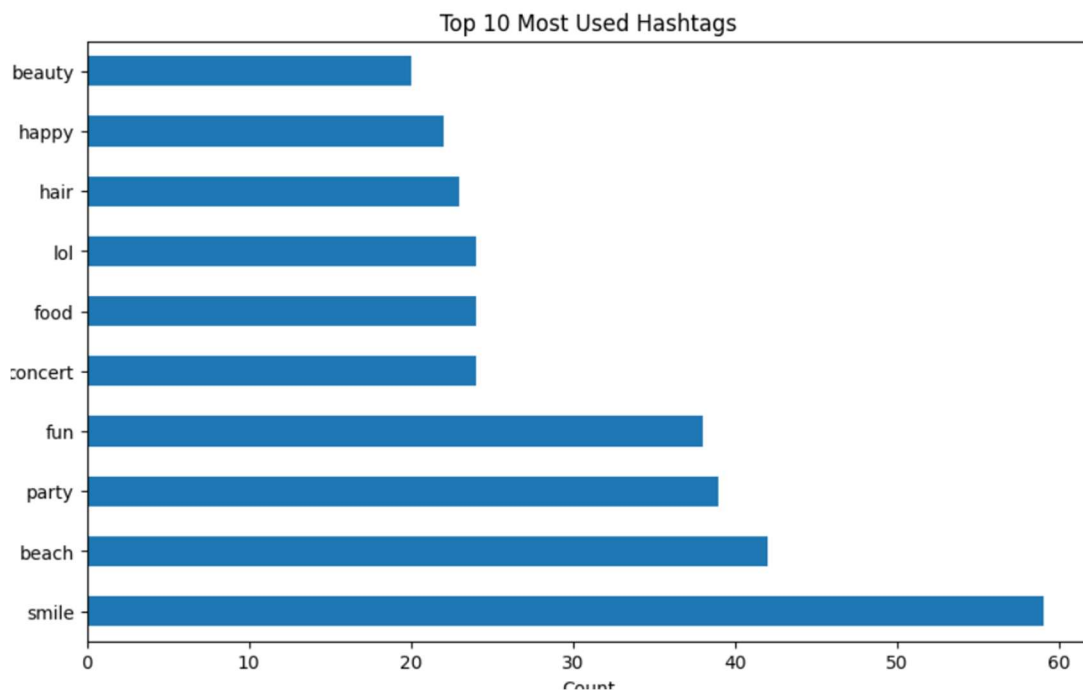
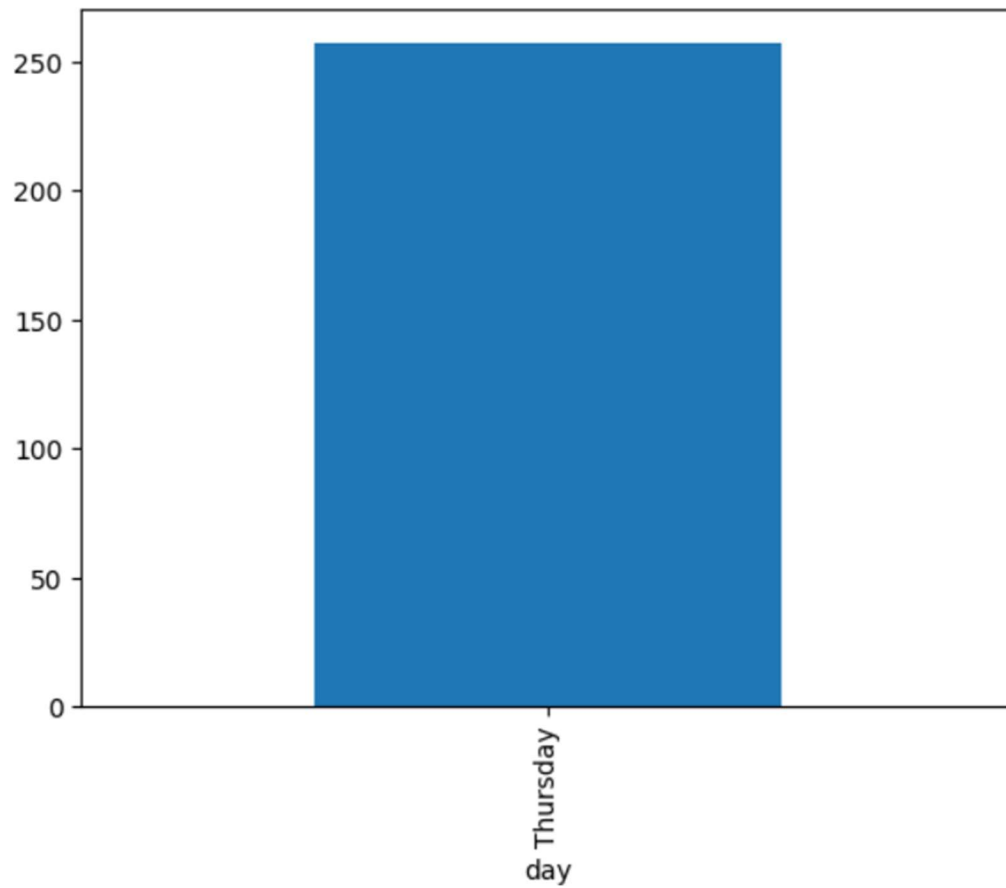
Expected Outcomes

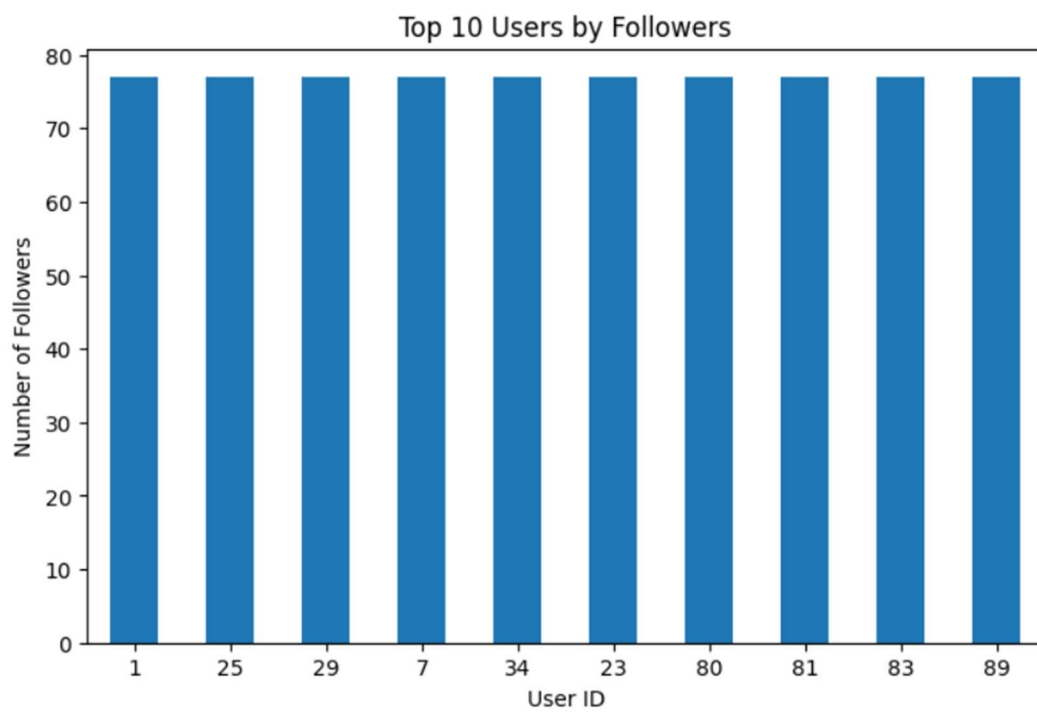
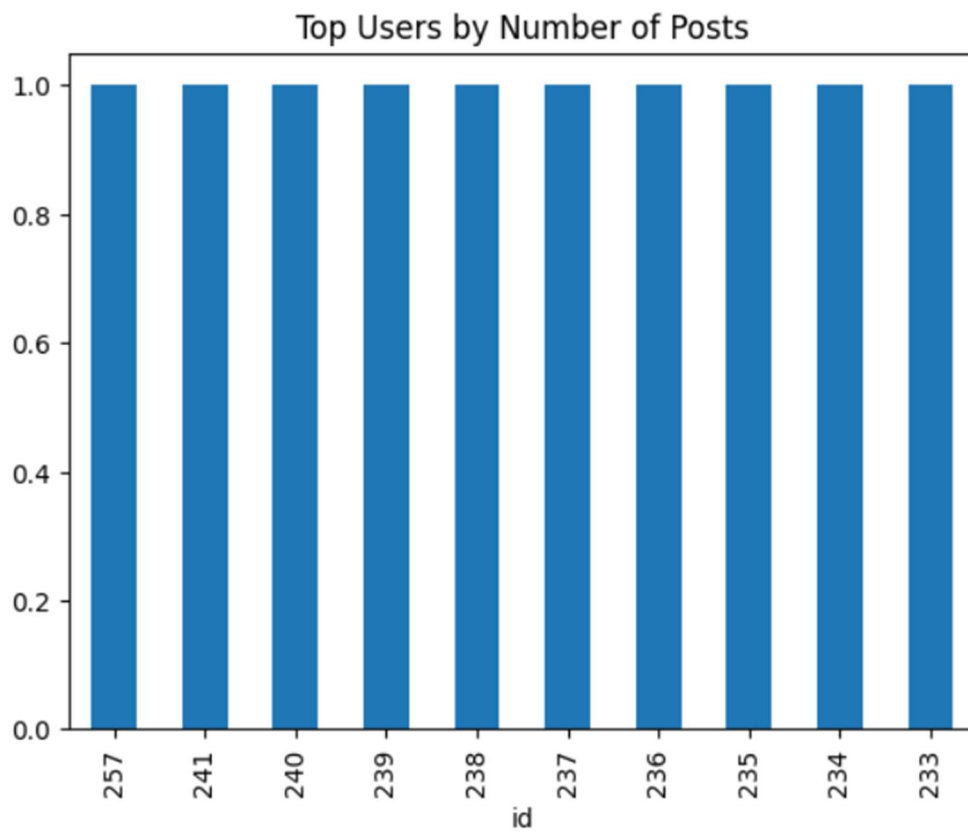
- Higher engagement rate**
- Increased follower growth**
- Improved brand visibility**
- Better audience retention**
- Stronger organic reach**

Screenshots & Charts

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Active vs Inactive Followers

